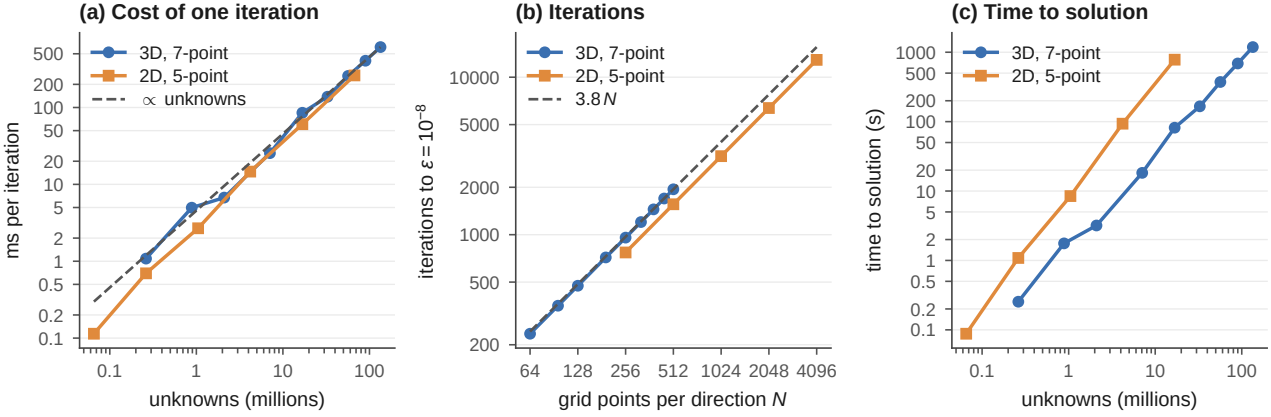
(a) Cost of one iteration
3D, 7-point
2D, 5-point
∝ unknowns
ms per iteration
500
200
100
50
20
10
5
2
1
0.5
0.2
0.1
0.1
1
10
100
unknowns (millions)
(b) Iterations
3D, 7-point
2D, 5-point
3.8N
iterations to ε = 10^−8
10000
5000
2000
1000
500
200
64
128
256
512
1024
2048
4096
grid points per direction N
(c) Time to solution
3D, 7-point
2D, 5-point
time to solution (s)
1000
500
200
100
50
20
10
5
2
1
0.5
0.2
0.1
0.1
1
10
100
unknowns (millions)